Supplementary file 2. Contribution decision, analysis plan, and amendments A1–A7

Retracted, replaced, or corrected: Checking the reference lists of 1,000 systematic reviews against Retraction Watch, PubMed, and Crossref

This file reproduces, as written, the two planning documents of 7 October 2026 and the post hoc amendments added on 24 and 25 October 2026. The plan was an internal document and was not registered. Its amendments A1–A3 were added at 10:32–10:36 UTC on 7 October, after the first build, and no hash of the plan's earlier text was kept, so its stated writing time is self-recorded. Amendments A4–A7 and addendum A7.1 were written after every original result was known; each file's SHA-256 was appended to a time-stamped project ledger, and the hashes below were checked against that ledger when this file was built.

For anonymized review, the author's name is replaced by "the author" and the affiliation in one software User-Agent string by "[affiliation]". Nothing else is changed. Each amendment states its own status when it was written, including whether the author had reviewed it at that time. Paths such as docs/, outputs/, and validation/ refer to the project archive; paths beginning WORK/deep refer to the working records of the October 2026 correction.

Document	Time (UTC)	SHA-256 of the file as written
Contribution decision	7 October 2026; file last saved 09:45:15 UTC	dec10a4a1a165c81fe1a92b4129e574855e08932a50c58b74f0d01832855dc95
Analysis plan, with amendments A1–A3	7 October 2026; writing time self-recorded as 09:44–09:55 UTC; file last saved 10:36:55 UTC	ee283b53a94bd3aec9900f54cb7c3c4315a84e6a943d47f7c4e4b44b2309d0f8
Amendment A4 (post hoc)	ledger entry 2026-10-24 17:11:27	cdb2ce915bcabe390304502867f3a7d5610add62838e41a1ec2bf0a3bd613ac8
Amendment A5 (post hoc)	ledger entry 2026-10-24 17:49:47	e72cd04c37cc5bb174fdce0ff38d1fd514eaf3302de6d162cf730828975caa91
Amendment A6 (post hoc)	ledger entry 2026-10-24 18:51:50	d50c3531e9af1e91d44ba477c04d106dacd0fd100d6427e3111fb38501767948
Amendment A7 (post hoc)	ledger entry 2026-10-24 22:06:47	8d4d059736f403ff859ece783c47d08a8709dddb6ee098913eaf40511ba52c98
Addendum A7.1 (post hoc)	ledger entry 2026-10-25 00:30:16	873d973998eff4884e12daec3f70948624bd0523858d389497b9bdc281e214c7

Contribution decision

Written 2026-10-07T09:44Z, before the review corpus was retrieved and before any reference was linked to a status source. The only data seen at this point were the Retraction Watch field distribution (record counts by `RetractionNature`, checked while validating the download) and the size of the PubMed frame (7,993 records).

Who could use the output

A research analyst or information specialist who maintains the evidence table behind a systematic review or meta-analysis in cardiometabolic medicine. The same check is useful to review authors just before submission or when planning an update, and to editorial staff who run reference checks.

What decision it supports

Which cited studies need a person to look at their post-publication status (retraction, expression of concern, correction, reinstatement), and whether the notice was already public when the review first appeared online or was issued afterwards. The first case is a question about the review's search and screening; the second is an evidence-maintenance trigger. Neither case says anything about whether the review's conclusions hold.

Narrow unresolved question

Published work on retracted studies in evidence syntheses mostly starts from known retracted items and searches forward for citing reviews (Avenell et al. 2019; Kataoka et al. 2022; Brown et al. 2022; Xu et al. 2025), or screens citing papers with a tool built on Crossref and Retraction Watch data (Graña Possamai et al. 2025, using the Feet of Clay Detector). Reference managers such as Zotero match DOIs and PMIDs against Retraction Watch for retractions only. Separate work compares retraction indexing across databases (Sebo et al. 2026; Salami, McCumber and Schneider, preprint v4 2026).

What is less well measured is the backward, reference-list view that an analyst actually has: starting from a review's own reference list, how much of it can be checked at all, and how much the answer depends on (1) which identifiers are used, (2) which status source is consulted, and (3) how precise the dates are. The question for this project is:

> *In a fixed, deterministically drawn cohort of open-access cardiometabolic systematic reviews and meta-analyses, how do identifier coverage, status-source coverage and date precision change the set of cited studies that a reference audit flags, and how often is the timing of a flagged notice relative to the review determinate?*

Comparator

An exact-DOI-only match of each reference against the frozen Retraction Watch snapshot. This is the simplest check an analyst can run from a DOI list. The comparison arms are:

1. exact DOI only, Retraction Watch only;
2. verified expanded identifiers (DOI, PMID, PMCID and NCBI-verified DOI/PMID crosswalks), Retraction Watch only;
3. arm 2 plus PubMed's own update relations (`CommentsCorrections` and the "Retracted Publication" publication type).

Crossref and Retraction Watch are one integrated source since 2023, and PubMed relations are partly fed by the same publisher notices, so agreement between arm 3 sources is reported as overlap and never as independent confirmation.

Estimand

Reference-occurrence-level proportions in the sampled cohort (one row per review-reference occurrence), with review-level cluster bootstrap intervals. The target frame is PubMed records retrieved by the prespecified query: PubMed publication type Systematic Review or Meta-Analysis, diabetes, obesity or heart failure by MeSH or title, publication date 2015 through 2025, PubMed's "pubmed pmc open access" filter. Unique cited works and notices are reported as separate units.

What counts as a useful negative result

- If the expanded crosswalk links few or no additional flagged studies, DOI-only checking is adequate for this corpus and the remaining gap sits in references without any identifier. That is worth knowing.
- If no reference in the cohort links to a retraction, the coverage and chronology ledger is still the result. The disease scope and date window will not be changed after seeing the hit count.
- If most timing classifications are indeterminate, that is a finding about date precision in the sources, not a failure of the audit.

How the work is framed

This is not the first study of retracted work in systematic reviews, and the tool is not the first to match references against Retraction Watch. The work is framed as a reusable, status-type-aware and time-aware reference audit tool, with an empirically measured coverage gap in one clinical domain. Its specific additions are: a backward audit from genuine JATS reference lists; a ledger that keeps "no identifier", "endpoint unavailable", "ambiguous identifiers", "record not found" and "verified record with no recorded event" apart; separate handling of retraction, correction, expression of concern and reinstatement claims with each source's date; and interval-based timing that admits indeterminacy. It does not estimate effects on pooled results, does not judge any study or review, and makes no clinical claims.

Analysis plan

- Written: 2026-10-07T09:44Z to 2026-10-07T09:55Z (UTC), before the review corpus was retrieved, before any reference list was parsed, and before any reference was matched to a status source.
- Status: internal, timestamped plan kept with the project files. It is not a formal preregistration in a public registry.
- Later changes are recorded in the "Amendments" section at the end with their date and reason. Earlier text is not edited.

Prior exposure

- Retraction Watch snapshot: the file was downloaded and validated (row count, column names, `RetractionNature` counts: 67,530 Retraction, 3,730 Expression of concern, 1,541 Correction, 162 Reinstatement; 72,963 records).
- PubMed frame: the frame query was run once with `retmax=0`; it returned a count of 7,993.
- Literature: the closest published studies were read (see `docs/prior_art.csv` and `docs/literature_search.md`). Their findings, for example that a large share of reviews citing retracted trials appeared after the retraction, are known in general terms.
- No review XML, no reference list and no linkage output had been seen.

1. Question

How do identifier coverage, status-source coverage and date precision change which cited studies in cardiometabolic systematic reviews and meta-analyses are flagged for status review, and how often is the timing of a flagged notice determinate relative to the review's first public online date?

2. Data sources (frozen)

Source	Version pinned	Use
Crossref Retraction Watch CSV	GitLab `crossref/retraction-watch-data`, commit `fe345427b516f7d507d67db5e1119d738cc25455` ("Retraction Watch data for 2026-10-06"), SHA-256 `063df713...0db4e5`	Status claims (Retraction, Correction, Expression of concern, Reinstatement) with notice dates and identifiers
PubMed (E-utilities)	Responses retrieved 2026-10-07, cached and hashed	Frame query; review dates; cited-record metadata; `CommentsCorrections` relations; publication types
PMC Article Datasets (AWS, `pmc-oa-opendata`)	Per-article JSON and JATS XML retrieved 2026-10-07; MD5 from the JSON, SHA-256 computed locally	Review full text and genuine reference lists; license codes
PMC ID Converter API	Responses retrieved 2026-10-07, cached	PMID to PMCID mapping for the frame; PMCID to PMID for cited PMCIDs

The status audit is current as of the snapshot. It does not claim that every notice was knowable when the review appeared.

3. Frame, selection and cohort

Frame query, PubMed, run 2026-10-07:

```
("Diabetes Mellitus"[MeSH Terms] OR "Obesity"[MeSH Terms] OR "Heart Failure"[MeSH Terms]
 OR diabet*[Title] OR obes*[Title] OR "heart failure"[Title])
AND ("Systematic Review"[Publication Type] OR "Meta-Analysis"[Publication Type])
AND ("2015/01/01"[Date - Publication] : "2025/12/31"[Date - Publication])
AND "pubmed pmc open access"[Filter]
```

- The publication window is 2015 through 2025, the last complete calendar year.
- The open-access rule is PubMed's "pubmed pmc open access" filter, which the PMC Open Access Subset page documents. Membership is confirmed per article with the `is_pmc_openaccess` field of the PMC cloud JSON.
- The full frame of PMIDs is saved as `data/interim/frame_manifest.csv` with its SHA-256 before any status lookup.

- Selection: sort the frame by `SHA-256("retraction-aware-evidence|2026-10-07|" + PMID)` ascending and take the first 1,000. This is deterministic, depends only on the identifier, and does not depend on any status information.
- Selected articles are not replaced. Articles that cannot be retrieved (no PMCID, no cloud version, not in the open-access subset, download error) and articles whose XML has no parseable reference list are reported by reason.
- Version rule when the cloud bucket has more than one version: prefer a version with `is_pmc_openaccess = true` and `is_manuscript = false`; among those, take the highest version number; if none, take the highest open-access version.
- This cohort is a sample of one indexing-defined frame. It is not a census of cardiometabolic reviews. Selection bias toward open-access journals and toward records with publication-type indexing is reported as a limitation.

4. Units

1. Review article (PMCID).
2. Reference occurrence: one `<ref>` element in a review's reference list. A `<ref>` holding several citations (`<citation-alternatives>` excluded) yields one occurrence per `element-citation/mixed-citation/nlm-citation` child.
3. Unique cited work: a connected component over identifiers (normalized DOI, PMID, PMCID), joined by co-occurrence within one reference and by verified NCBI crosswalk records.
4. Notice: one Retraction Watch record (`Record ID`) or one PubMed notice record (PMID of the notice in `CommentsCorrections`).

Several reviews can cite one work and one work can have several notices. Counts of notices are never reported as counts of studies.

5. Identifier handling

- Keep the original string of every identifier with its provenance: `pub-id` tag, `ext-link/uri` link, or free text in the reference.
- DOI normalization is conservative: trim whitespace; remove a leading `doi:/DOI:` label or a `https://doi.org/`, `http://dx.doi.org/` style prefix; percent-decode; lowercase (DOI names are case-insensitive); strip trailing `.`, `,`, `;`, `:` and a trailing `)` or `]` only when it is unbalanced. The result must match `^10\.\d{4,9}/\S+$`; otherwise the DOI is recorded as malformed and not used.
- PMIDs must be positive integers; PMCIDs must match `PMC\d+`.
- A DOI that equals a Retraction Watch `RetractionDOI` (or a PMID that equals a `RetractionPubMedID`), or a PubMed record whose publication type is "Retraction of Publication", "Published Erratum" or "Expression of Concern", is a cited notice. It is reported separately and is never treated as the retracted article.

6. Linkage arms

- Arm A, exact DOI only: normalized reference DOIs against normalized Retraction Watch `OriginalPaperDOI`. No other identifier.
- Arm B, verified expanded identifiers: all reference identifiers (DOI, PMID, PMCID) plus NCBI-verified crosswalks (PMCID to PMID via the ID Converter; PMID to DOI from the PubMed record's own `ArticleIdList/ELocationID`; DOI to PMID by a PubMed search verified by exact DOI equality in the returned record) against Retraction Watch `OriginalPaperDOI` and `OriginalPaperPubMedID`.
- Arm C: arm B plus PubMed relations on the cited record: `RetractionIn`, `RetractedandRepublishedIn` (retraction), `ExpressionOfConcernIn` (expression of concern), `ErratumIn`, `CorrectedandRepublishedIn` (correction), `UpdateIn` (update, reported separately), and the "Retracted Publication" publication type (retraction without a linked notice).
- Reinstatement exists only in Retraction Watch. No reinstatement value is created for PubMed.
- Conflicts are kept: each claim is stored with its source, notice identifier and date. No source is preferred and no "most severe label" is chosen silently. Summaries that need one row per occurrence use "any retraction claim", "any expression of concern claim", and so on, as separate indicators.
- A title/author/year fuzzy matcher (normalized title similarity of at least 90 on `rapidfuzz.fuzz.token_sort_ratio`, title length at least 30 characters, cited year within 1 of the Retraction Watch original-paper year) produces a separately labeled review queue for occurrences with no exact link. It

is never counted as a confirmed match.

7. Coverage ledger (mutually exclusive, per occurrence, arm C)

1. `no_identifier`: no usable DOI, PMID or PMCID.
2. `malformed_identifier_only`: only malformed identifiers.
3. `endpoint_unavailable`: a required NCBI lookup failed after bounded retries. These are not counted as "no event".
4. `ambiguous`: identifiers point to more than one PubMed record (for example a tagged PMID and a DOI that resolves to a different PMID).
5. `pmid_not_found`: a PMID with no PubMed record and no usable DOI.
6. `doi_not_in_pubmed`: a DOI with no PubMed record; checked against Retraction Watch by DOI only.
7. `verified_pubmed_record`: one PubMed record; checked against both sources.

Within categories 6 and 7, status is `event_recorded` (with type) or `no_recorded_event_as_of_snapshot`. None of these categories says a study is sound.

8. Chronology

- Review first public online date: JATS `pub-date` with `pub-type="epub"` or `date-type="pub"` and `publication-format="electronic"`; else PubMed `ArticleDate DateType="Electronic"`; else PubMed journal-issue date. Issue date: JATS `ppub/collection` or PubMed issue date.
- Every date is an interval. A full date is one day; year-month spans the month; year only spans the year; a missing date is unknown.
- Notice dates: Retraction Watch `RetractionDate`; PubMed notice date from the notice record (electronic article date, else issue date). A PubMed relation without a notice PMID or a fetchable notice record has an unknown date.
- The acceptance date is not used as the date when references were finalized.
- Timing per claim: `before_review` when the notice interval ends before the review interval starts; `after_review` when the notice interval starts after the review interval ends; otherwise `indeterminate`.

9. Primary outputs

1. Reference-level identifier coverage: share of occurrences with any usable identifier, and by identifier type and provenance.
2. Uniquely resolved status matches: occurrences and unique works with at least one claim, by claim type, in arms A, B and C.
3. Newly linked matches from authoritative crosswalks (in B, not in A) and from PubMed relations (in C, not in B), with source overlap reported as overlap.
4. Share of flagged occurrences whose timing is determinate, and the split before/after.
5. The coverage ledger counts.

Proportions are reported with review-level cluster bootstrap 95% percentile intervals (2,000 resamples of reviews with replacement, seed 20261007). Counts that are small are reported as counts with no interval.

10. Sensitivity analyses (fixed set)

- S1: tagged identifiers only (drop identifiers mined from free text and links).
- S2: timing against the issue date instead of the first online date.
- S3: the six-month rule used by Brown et al. (2022): a claim counts as "before review" only if the notice is at least 183 days before the review's online date.
- S4: unique-work level instead of occurrence level.
- S5: excluding occurrences that cite a notice rather than an article.

11. Missing data and exclusions

No imputation. Unavailable, missing, not found, ambiguous and not evaluated are separate states. Failed requests are retried with backoff (up to 5 attempts, batch splitting down to single identifiers); identifiers that still fail are `endpoint_unavailable`.

12. Validation

- Software tests on controlled fixtures: DOI punctuation, URL prefixes, duplicated notices, notice DOI swaps, multiple PMID relationships, incomplete dates, failed requests. Fixtures are labeled and never mixed with study rows.
- Source reconciliation: every exact match is re-checked by a separate routine that re-reads the raw Retraction Watch CSV row and the raw cached PubMed XML and confirms identifier equality, claim type and date.
- Determinism: deterministic stages are re-run from cached inputs and substantive outputs are compared with tolerance 0 for counts and 1e-12 for proportions.
- Human review of flagged rows and a stratified sample of reference parses is described separately and is not part of the automated checks.

13. What will not be done

No inference about whether a review's pooled effect changes, no judgment about whether a citation endorses or criticizes a study, no misconduct attribution, no change of disease scope or date window after seeing results.

Amendments

A1 (2026-10-07T10:32Z, after the first build): exploratory notice dates from PubMed `RefSource`

- Observed: 784 PubMed status relations (783 `ErratumIn`, 1 `RetractionIn`) carry no notice PMID, so under section 8 their notice date is unknown and their timing is indeterminate. Each relation still has PubMed's `RefSource` citation string for the notice (for example "Diabetes Care. 2014 Mar;37(3):887"), which contains a year and usually a month.
- Change: an exploratory table parses the year, month and day in `RefSource` into a date interval with the same rules as section 8 and recomputes timing for those relations only. It is labeled post hoc everywhere it appears.
- Not changed: the primary timing results, which keep these relations as indeterminate exactly as planned.
- Reason: the result shows how much of the indeterminate share comes from a missing notice PMID rather than from overlapping dates. It was decided after seeing the count of undated relations and before looking at the reparsed timing.

A3 (2026-10-07T10:36Z): current NLM name of the notice publication type

- Section 5 named the PubMed notice type "Retraction of Publication". The current NLM publication type list names it "Retraction Notice"; every retraction notice record retrieved here carries the new name, and none carries the old one. The implementation now recognizes both names. Found while reconciling the first run's cited-notice table: 3 references to "Retraction Notice" records had not been labeled as cited notices.
- Effect: only the cited-notice table and sensitivity analysis S5 can change. Status claims do not depend on the notice's own publication type.

A2 (2026-10-07T10:32Z): source-disagreement rule made explicit

- Section 8 classifies each claim. For occurrence-level timing tables, an occurrence with several claims of the same type (for example one from each source) is called determinate only when all of those claims agree; otherwise it is indeterminate. This rule was applied in the first analysis run and is recorded here because section 8 did not state it.

Amendment A4: checking each identifier link against the matched record

- Written: 2026-10-24, between 17:09Z and 17:11Z (UTC). The file's SHA-256 is recorded in `validation/ledger_2026-10-24.jsonl`, in the line appended after this file was finished; a file cannot contain its own hash.
- Status: post hoc. Every result of the study was known when this was written, including the manuscript submitted on 18 October 2026 and the item-level AI check of 24 October 2026 (`validation/item_verification_2026-10-24.xlsx`, SHA-256 `e3ead296c11e4788c94539f4195cdb8228460113dffa9237d444cf8e42760d00`).
- Drafted by an AI agent (Claude Opus 5.5, claude-opus-5-5, via Claude Code) for the author. The author has not yet reviewed it; the review is queued in the deep-pass author review list.
- `docs/analysis_plan.md` and its amendments A1 to A3 are not edited. This file adds to them.

1. What was wrong

Section 6 of the plan accepted any exact identifier match as a link. The AI check of 24 October went back to the source records for all 120 flagged claim rows and found four rows, in three references, where an identifier joined a reference to the wrong record. Applying the rules below to every link in the study removed four more links: the PubMed half of the erratum case in item 3 and three further references (the last two in item 3, and item 4).

1. **A supplement-level DOI.** Review PMC8518965, reference 39, cites Kramná et al., "Mass sequencing of the faecal virome: a study in children with early onset of islet autoimmunity", an abstract in *Diabetologia* 2014;57(suppl 1):S78. The DOI printed with it, 10.1007/s00125-014-3355-0, belongs to the whole EASD abstract supplement. Retraction Watch record 6537 uses the same DOI for a different abstract in that supplement, "Evaluation of DiaPep277® treatment in type 1 diabetes by integrated analysis" (DIA-AID group; retraction notice 10.1007/s00125-015-3542-7, reason "Misconduct by Third Party"). The two titles share no subject, the author lists do not overlap, and the title similarity is 42.7 on the scale used below. The pipeline counted the reference as citing a retracted study. It does not. (Worksheet row FC067.)
2. **A DOI printed under a paper of the same title.** Review PMC11983615, reference B106, cites Ashtagi et al., *Int J Intell Syst Appl Eng* 2023;11:714–26, but prints DOI 10.1155/2022/2389636. That DOI is Padhy et al., *Comput Intell Neurosci* 2022 (PubMed 35634091), which carries the same title and was retracted in 2023 (PubMed notice 37564530, Retraction Watch record 48254). Title matching cannot separate the two; the authors and journals can. (Rows FC031 and FC032.)
3. **A reference that cites an erratum but carries the article's PMID.** Review PMC9630197, reference CR8, cites the 2018 *Diabetes Care* erratum to the PREDIMED-Reus trial (DOI 10.2337/dc18-er10, PubMed 30104300) with the trial's PMID (20929998). Both the Retraction Watch correction record 19033 and the PubMed ErratumIn relation were then attached to the reference, so the erratum was reported as having a correction. Section 5 of the plan says a cited notice "is never treated as the retracted article". The SQL applied that only when every identifier in the reference pointed to the notice. (Row FC121.) Two more references of the same kind were found when the rule was run on every link: PMC10513216 reference ref069 ("Erratum: the National Institutes of Health and guidance for reporting preclinical research", PubMed relation to notice 25889769) and PMC11172189 reference B67 ("Correction: Plasma Kynurenines and Prognosis in Patients with Heart Failure", notice 32109260).
4. **A book cited with the identifiers of an article.** Review PMC10836683, reference ref113, cites the Routledge book *Physical activity in low- and middle-income countries* (Siefken et al., 2021) but carries the DOI and PMID of a 2021 *Journal of Physical Activity and Health* commentary by the same first author (10.1123/jpah.2021-0576, PubMed 34649222), which has an erratum (PubMed 37775101). The erratum belongs to the commentary, not to the book. This case was outside the 120 checked rows; the rule found it.
5. **Season dates cut to their first month.** PubMed and JATS give some issue dates as a span ("Jan-Feb"). `rae.dates.from_parts` read only the first month, so the interval ended a month early. For the retraction notice PubMed 33563391 (row FC061) the true interval, 1 January to 28 February 2021, overlaps the citing review's online date of 5 February 2021; the PubMed claim was classed "before review" and is indeterminate. The same fix changes the issue-date interval of 8 notice records, 1,627 cited records, 14 PubMed review records and 21 JATS review dates (no review's timing class changes). Only two of the notices lack an electronic date, so only those two reach a notice interval, and only FC061 changes class.

6. **Authors tagged as** `<string-name>`**.** `rae.jats._first_author` read `<name>` and `<collab>` only. The AI check found 15 of 100 sampled references with a missing or wrong first author for this reason. The field is shown in the review queues and is not used for linkage, ledger classes, claims or counts.

The check also left two rows unchanged: FC119 (Retraction Watch gives the reinstatement date as unknown; the pipeline's date stands and is reported with that caveat) and FC039 (an undated PubMed retraction relation, indeterminate as section 8 requires).

2. Rules added

Rule A4.1: bibliographic agreement of each link

Applied to every link from a reference to a Retraction Watch record (by DOI or PMID, in arms A, B and C) and to every PubMed record whose relations or publication type give a status claim (arm C), in the primary analysis and in the tagged-only sensitivity analysis S1, and in `rae audit`. "Crossref" identifiers enter the study only as DOIs; there is no separate Crossref status source, so the rule covers every DOI match. Three checks compare the matched record with the reference as printed:

- **Title.** The record's title, normalized as in the title queue, against the reference text (RapidFuzz `partial_ratio`) and against the parsed article title (`token_sort_ratio`); the higher score counts. A score below 70 disagrees. The check is skipped when the record has no usable title ("[Not Available]"), when a PubMed title is in square brackets (NLM's mark for a translated title, which cannot match a reference printed in the original language), or when the reference has no text.
- **First author.** Some word of two or more letters from the record's first author appears as a word in the reference text. Skipped for group authors and empty names.
- **Source.** The record's journal against the reference's tagged source, allowing abbreviations (each word of the shorter form starts a word of the longer form, in order). Skipped when either is empty.

A link is rejected when two of the three checks disagree. One disagreement alone is common in correct links, so such links are kept and listed for a person to look at (`outputs/tables/amendment_a4_review_list.csv`, 65 links; typical causes are guideline titles shortened in the reference, renamed journals and author names written in another order).

Why 70. The 116 links that the 24 October AI check confirmed as the right work all score 84.6 or more; the Kramná link scores 42.7. Seventy sits well inside that gap. On all 2,522 links in the study, any threshold from 58.8 to 84.5 gives exactly the same decisions, and the three retraction links rejected are the same for every threshold from 45 to 95 (`outputs/tables/amendment_a4_threshold_sensitivity.csv`). The threshold was chosen after the results were known, which is why the whole range is reported.

Why two of three. A single check fails on correct links for ordinary reasons. Requiring two different kinds of disagreement (title and author for the supplement DOI; author and journal for the same-title paper) rejects the known false links without touching any link that the AI check confirmed.

Rule A4.2: references that cite a notice

A claim is dropped when the reference cites the notice itself: a DOI or PMID printed in the reference is an identifier of the claim's notice and not also an identifier of the article, and the reference's title is a notice title (it begins "Erratum", "Correction", "Corrigendum", "Retraction note", "Expression of concern", "Withdrawal" or similar). This applies section 5 of the plan to mixed references. A reference that gives the article's title with one identifier of its notice still cites the article, and its claim is kept. In this study 28 links from 25 references printed an identifier of their notice; the 4 links from 3 references with a notice title were dropped and the rest kept (all of the rest carry the article's title, and most were already labeled ambiguous in the ledger).

Implementation corrections that follow section 8 and the parser's stated purpose

- A month span written as a season ("Jan-Feb", "Dec-Jan") is read as an interval from the first day of the first month to the last day of the second (`rae.dates.from_parts`). Named seasons ("Spring") still give the whole year.
- `<string-name>` is read as an author (`rae.jats._first_author`).

Rejected claims are not deleted. They are kept with the rule that removed them in `data/processed/a4_claims_rejected.parquet`, in `outputs/tables/amendment_a4_rejected_links.csv` and on the sheet "A4 rejected links" of the review queue workbook.

3. Code and tests

- `src/rae/guard.py` (new), called from `rae.build.run_variant`; `src/rae/dates.py`; `src/rae/jats.py`; tables in `src/rae/analysis.py`, `src/rae/outputs.py` and `src/rae/numbers.py`.
- `tests/test_guard.py` and `tests/fixtures/fixture_rw_a4.csv`: a synthetic end-to-end fixture with the supplement-DOI case, the same-title case, the cited-erratum case and two correct controls, run through the study SQL and the guard; a regression check on the published strings of the Kramná and Ashtagi references; season spans; `<string-name>`. The full suite passes (64 tests, 2026-10-24).
- `rae validate` no longer rewrites `validation/human_validation_worksheet.xlsx` when it exists. The worksheet was handed out on 7 October and is a record.

4. What it changes

Everything was regenerated offline from the cached 7 October responses and the pinned Retraction Watch file, with no new downloads: `rae acquire` with `RAE_OFFLINE=1` (re-parsing only), then `rae build`, `rae analyze` and `rae validate` (all checks passed; the offline rerun of build and analyze matched every output). The pre-amendment state is kept in `_superseded/pre_A4_2026-10-24/`.

Measure (arm C unless stated)	Before A4	After A4
Identifier links checked	–	2,522
Links rejected (A4.1 / A4.2)	–	8 (4 / 4)
Retraction-flagged occurrences	46	44
Reviews with a retraction flag	41 (4.1%)	40 (4.0%)
Unique works with a retraction claim	44	42
Retraction: both sources / PubMed only / Retraction Watch only	35 / 7 / 4	34 / 7 / 3
Retraction timing: before / after / indeterminate	16 / 28 / 2	14 / 28 / 2
Retraction-flagged, arm A / arm B	39 / 39	37 / 37
Correction-flagged occurrences, arm B / arm C	7 / 2,169	6 / 2,165
Any claim, arm A / B / C	55 / 57 / 2,266	53 / 54 / 2,260
Reviews with any claim (arm C)	808	807
Links added by the crosswalk (B not A)	2	1
Correction timing determinate	64.1% (62.0–66.1)	64.0% (62.0–66.0)
Title-queue candidates	0	1

The Kramná case alone accounts for the change from 46 to 45 occurrences, 44 to 43 works, 4 to 3 Retraction Watch-only and 16 to 15 before review; the Ashtagi case accounts for the rest of the retraction changes (`deep/S10/a4/a4_retraction_scenarios.csv` in the deep-pass folder). Expressions of concern, the reinstatement, the update claims, identifier coverage, the coverage ledger, the 48,691 unique works, the cited-notice table and the amendment A1 table do not change. Occurrence-level timing classes do not change through the date fix; the claim-level table by source does (PubMed retraction claims before review 15 to 13, indeterminate 1 to 2). The one title-queue candidate is the Ashtagi reference against the Padhy record, which is what the queue is for. A full list of old and new values, with their source files, is kept with the deep-pass records (`corrected_numbers.md`).

5. Limits

- The rules were written after the false links were seen and are tuned to catch them. On the 116 confirmed links they reject nothing, but the 2,402 links outside the 120 AI-checked rows, nearly all of them corrections, have not been checked another way; the rule found one false link among them (the Siefken book) and lists 65 more for a person to look at.
- The rule checks only links that produced a status claim. A wrong identifier on a reference with no recorded notice is not looked for, so the coverage ledger can still contain references whose identifier points to another work.
- A false link whose record shares the reference's title and first author and journal would pass. None was found in the checked rows.
- An independent re-computation of these numbers by a separate agent has not yet been done.

Amendment A5: retraction claims by notice type and stated reason

- Written: 2026-10-24, between 17:48Z and 17:50Z (UTC). The file's SHA-256 is recorded in `validation/ledger_2026-10-24.jsonl`, in the line appended after this file was finished.
- Status: post hoc and descriptive. When this was written, every result of the study was known: the manuscript submitted on 18 October 2026, the item-level AI check of 24 October 2026 (`validation/item_verification_2026-10-24.xlsx`), amendment A4 and its rerun (`docs/amendment_A4_2026-10-24.md`, SHA-256 `cdb2ce91…13ac8`), the AI re-computation of the A4 numbers (`validation/ai_check_recount_2026-10-24.json`, SHA-256 `ae1c615f…08c9d`), and a first run of the code described below. A research note written before A4 had already predicted three of the figures (7 retract-and-replace occurrences, 11 Hindawi occurrences, 8 occurrences in 7 reviews); section 4 explains why the amended counts differ.
- Drafted by an AI agent (Claude Opus 5.5, claude-opus-5-5, via Claude Code) for the author. The author has not yet reviewed it; the review is queued in the deep-pass author review list.
- `docs/analysis_plan.md`, `config/config.yaml` and amendments A1 to A4 are not edited. No claim, link, timing class or table written by `rae analyze` changes.

1. Why

The main analysis counts a reference as retraction-flagged whenever a status source records a retraction for the cited work. That single count mixes two different situations. Some notices withdraw a paper whose findings should no longer be used. Others retract a paper and publish a corrected version of the same work in its place (the 2013 PREDIMED trial in the *New England Journal of Medicine* is the best-known case). A review that cites the original PREDIMED paper after June 2018 cites a superseded version, not a discredited trial. Readers of the paper need to know how many of the flags are of each kind, why the other retractions happened, and how many notices were old enough that a review team could reasonably have found them.

2. Definitions

The input is the amended arm C claim table (`data/processed/claims.parquet` after A4, SHA-256 `1f5eb8e1…299b7d`): 44 retraction-flagged reference occurrences in 40 reviews.

- **Notice type.** An occurrence is `retract_and_replace` when a Retraction Watch record linked to the cited work has "Retract and Replace" in its Reason field, or PubMed records the relation `RetractedandRepublishedIn` on the cited record. The type belongs to the work, so it applies to every occurrence of that work. All other occurrences are `retraction`. Only what the sources record is used. Titles, page order and notice wording are not used to infer a type.
- **Reason groups.** Each term in the Retraction Watch Reason vocabulary (112 terms in the pinned 6 October 2026 snapshot) is mapped to one of nine groups, or marked as a process term. Process terms say who investigated, that the authors did not respond, or that an earlier notice was upgraded; they do not give a reason and form no group. The full map is in `src/rae/amendment_a5.py` (`REASON_MAP`) and in `outputs/tables/amendment_a5_reason_map.csv`. A test fails if the snapshot contains a term the map does not cover.
- **Primary group.** A record usually lists several reasons. The primary group is the first one present in this order: replacement; paper mill, compromised peer review or generated content; fabrication, falsification or a misconduct finding; plagiarism or duplicate publication; image problems; ethics, consent or approval; error; concerns about data or results with no stated cause; publisher or administrative reasons. The order runs from the reason that most directly concerns whether the work can be trusted to the least. Because any such order is a choice, the non-exclusive count of every group is reported too.
- **Coarse reason.** The groups from paper mill to ethics are combined as "misconduct, manipulation or duplication". The remaining coarse classes are replacement, error, concerns with no stated cause, administrative, and "not recorded" for occurrences that only PubMed flags. PubMed does not code reasons.
- **Hindawi.** Any DOI of the cited work starts with 10.1155/. The DOI prefix is used because Retraction Watch lists the publisher of some of these journals as Wiley, which acquired Hindawi.
- **Timing.** This is the occurrence-level class of the main analysis. It is determinate only when every retraction claim on the occurrence agrees. The primary rule compares the notice date with the review's first online date. The six-month rule (sensitivity analysis S3) counts a notice as before the review only when it came at

least 183 days before the review went online.

- **Work.** The pipeline's `work_id`. One reference (PMC7551578 B59, PREDIMED) has identifiers that point to more than one record, so it has no `work_id` in the ledger. Here it is assigned to PubMed 23432189 because all its retraction claims name that record. The work count is 42 with or without it.

3. Results (`outputs/tables/amendment_a5_*.csv`)

Measure	Occurrences	Reviews	Works
Retraction-flagged (arm C, after A4)	44	40	42
Retract and replace	7	7	5
Other retractions	37	34	37
Hindawi DOI (10.1155)	10	10	10
Other retraction, notice at least 183 days before the review	7	7	7
Other retraction, notice before the review (primary rule)	10	9	10

- The five replaced works are PREDIMED (NEJM 2013, cited twice), Cochrane review CD001872.pub2 (2009, cited twice), Karalunas et al. (JAMA Psychiatry 2014), the UK Biobank mental health paper (BJPsych Open 2018) and the 2021 *Lancet* meta-analysis of obesity pharmacotherapy. Four of the seven occurrences were in reviews that went online after the notice: both PREDIMED citations, the JAMA Psychiatry paper and one Cochrane citation.
- For the Cochrane review, only Retraction Watch records a replacement ("Retract and Replace; Removed"). PubMed links it to the updated review only as UpdateIn. Both occurrences are Retraction Watch-only, and both are counted here as retract and replace.
- **Coarse reason of the 37 other retractions:** misconduct, manipulation or duplication 18; concerns about data or results with no stated cause 9; error 3; not recorded (PubMed only) 7.
- **Primary groups:** paper mill or peer review 12; plagiarism or duplicate publication 5; image problems 1; error 3; concerns with no stated cause 9. Seven occurrences carry the term "Paper Mill" itself.
- **Timing of the 37 under the six-month rule:**
  - 7 notices at least 183 days before the review went online;
  - 3 within the 183-day window;
  - 25 after the review went online;
  - 2 indeterminate.
- **The seven older notices.** Each sits in a different review, 389 to 4,508 days before the review went online:
  - Family Practice 2009 (duplicate publication);
  - British Journal of Nutrition 2013 (error);
  - British Journal of Clinical Pharmacology 2015 (compromised peer review and ethics);
  - Journal of Cellular Physiology 2018 (error in methods);
  - Diabetology & Metabolic Syndrome 2022 (duplicate publication);
  - two 2022 papers in the Hindawi *Journal of Healthcare Engineering* (paper mill or peer review).
- **Hindawi.** Of the 10 Hindawi occurrences, 7 were cited in reviews that went online before the notice, and five of those seven notices are dated 2023. Two notices were at least 183 days old and one was within the window.

The work-level table (`outputs/tables/amendment_a5_retracted_works.csv` and the sheet "Retracted works" in `outputs/workbooks/amendment_a5_retracted_works.xlsx`) has one row for each of the 42 works. Each row gives the DOI, PMID, journal, publication year, notice type, Retraction Watch record and notice date, PubMed notice and date, reason groups with the raw reason string, the number of citing reviews, and how many citing occurrences came after the notice, before it, or could not be placed.

4. Relation to the earlier predictions

The research note's figures (7 of 45, 11 Hindawi, 8 occurrences in 7 reviews) were made after removing only the Kramná supplement-DOI link (FC067), before A4. Run on the pre-A4 claims minus that one occurrence, this code gives exactly those figures: 45 occurrences, 7 retract and replace, 11 Hindawi, 8 occurrences in 7 reviews. A4 also removed the Ashtagi reference (PMC11983615 B106). That reference carried the DOI of a retracted Hindawi paper with the same title (10.1155/2022/2389636), and its review went online 603 days after that paper's notice. Removing it lowers each figure by one where it counted: 44 occurrences, 10 Hindawi, and 7

occurrences in 7 reviews. The retract-and-replace count does not change. The scenario output is in the deep-pass folder (`deep/S10/a5/a5_crosscheck_result.json`).

5. Limits

- Retraction Watch reasons are that database's coding of the notice, not findings of this study. "Concerns about data" often means an investigation that did not end in a stated finding.
- Seven of the 37 other retractions are flagged only by PubMed and have no recorded reason. They are reported as "not recorded", not assigned a group.
- One PubMed retraction relation has no notice record: the 2001 ACOG Practice Bulletin on gestational diabetes, cited in PMC4620826. Its RefSource dates the notice to August 2013, which is 783 days before that review (amendment A1, exploratory). The titles and page order suggest a guideline withdrawn when its successor (Practice Bulletin No. 137) appeared, but no source records it as a replacement, so it stays a retraction with indeterminate timing. Dated by A1 and counted as a retraction, it would make the older-notice figure 8 occurrences in 8 reviews. Treated as a replacement, the figure stays 7 in 7.
- The split says nothing about whether a review used the retracted study in its synthesis. That needs reading the reviews and is not part of this amendment.

6. Code and how to rerun

- `src/rae/amendment_a5.py` (new) reads the processed files and writes `outputs/tables/amendment_a5_retraction_occurrences.csv`, `amendment_a5_retracted_works.csv`, `amendment_a5_summary.csv`, `amendment_a5_reason_map.csv` and `outputs/workbooks/amendment_a5_retracted_works.xlsx`. It writes nothing else.
- `src/rae/cli.py` has a new stage, `rae a5`, which is run after `rae analyze`. It is not part of `rae all`, so the outputs that `rae validate` checks are unchanged.
- `tests/test_amendment_a5.py` (new, 9 tests) covers:
  - splitting reason strings, group priority and process terms;
  - a failure on unmapped terms;
  - notice type from each source, and that a plain RepublishedIn relation or an upgraded notice is not a replacement;
  - the timing rule;
  - a synthetic end-to-end case with an ambiguous reference and a lead-window notice;
  - map coverage of the pinned snapshot vocabulary;
  - a regression check of the study counts above.

The full suite passes (73 tests, 2026-10-24).

- Two runs of `rae a5` gave byte-identical CSV files. The workbook differs only in its save timestamp (`docProps/core.xml`).

Amendment A6: Crossref publisher deposits as a third status source

- Written: 2026-10-24, between 18:49Z and 18:51Z (UTC). The file's SHA-256 is recorded in `validation/ledger_2026-10-24.jsonl`, in the line appended after this file was finished.
- Status: post hoc and descriptive. Every result of the study was known when this was written: the manuscript submitted on 18 October 2026, the item-level AI check of 24 October 2026, amendment A4 and its rerun, and amendment A5. The Crossref records were retrieved and the analysis was run before this text was written. `src/rae/crossref_status.py` (retrieval, link checks including the notice-subject check, snapshot rule) was last modified at 17:46:34Z, before the first Crossref request at 17:46:46Z. The comparison code in `src/rae/crossref_analysis.py` was written while the records were downloading; its reason codes, the Retraction Watch title search, the date flags and the timing column were refined after the first results were seen.
- Drafted by an AI agent (Claude Opus 5.5, claude-opus-5-5, via Claude Code) for the author. The author has not yet reviewed it; the review is queued in the deep-pass author review list.
- `docs/analysis_plan.md`, `config/config.yaml` and amendments A1 to A5 are not edited. No claim, link, ledger class, timing class or table written by `rae build` or `rae analyze` changes. A6 adds a comparison beside them.

1. Why

The study reads status from two places: the pinned Retraction Watch file (data for 2026-10-06) and PubMed records retrieved on 2026-10-07. A4 noted that Crossref entered the study only as a source of DOIs. Yet when a publisher registers a retraction, correction or expression of concern at Crossref, it can deposit an `update-to` relation on the notice's DOI record naming the article it changes. Crossref shows the same relation on the article as `updated-by`, and since 2023 it also lists Retraction Watch entries there, each marked with its source. Publisher deposits are a third record, kept by a different party for a different purpose. Reference managers and some retraction checkers read them. The question here is how far that third record agrees with the other two, and why it does not when it does not.

2. Method

Frame. All 48,691 unique cited works of the study. 47,394 have at least one DOI (47,425 distinct DOIs, taken from the work's identifier nodes: printed, from PubMed records or from the PMC ID converter). Works without a DOI cannot be looked up and are left out; one retraction-flagged work and 19 correction-flagged works fall there. The planned fallback (limit the lookups to the 42 retraction-flagged works and every work flagged by any source if the full set took too long) was not needed.

Retrieval. Crossref REST API, public pool, User-Agent `[affiliation]-deep-pass/1.0`, no contact address, one request per 1.05 s (Crossref returned `x-rate-limit-limit: 1` per second). DOIs were sent 60 at a time as `filter=doi:…` queries; DOIs that a batch did not return, or that contain a comma, were looked up one by one at `/works/{doi}`, where HTTP 404 means Crossref holds no record. Then every notice DOI named in a publisher or Retraction Watch entry, every DOI of a PubMed notice behind a study claim (re-read offline from the 7 October cache) and every Retraction Watch notice DOI behind a study claim was fetched the same way. 1,445 requests ran between 17:46:46Z and 18:13:29Z on 24 October (847 answered 200, 568 answered 404, 30 were rate-limited with HTTP 429 and retried). Every response is cached with its SHA-256 under `data/cache/crossref_2026-10-24/`, and `request_log.jsonl` there lists each one.

What counts as a Crossref flag. A `updated-by` entry whose `source` is `publisher` and whose type is retraction, partial retraction, withdrawal or removal (counted as retraction), expression of concern, or correction, erratum or corrigendum (counted as correction). Types `new_version` (35 entries), `addendum` (4), `new_edition` (1) and the non-standard `err` (2) were not counted. Entries with `source: retraction-watch` were not counted as Crossref flags; they are used only to explain differences.

Link checks. Each pair of a reference and a Crossref update on its work's DOI went through the A4 rule (title, first author and journal of the Crossref record against the reference; rejected at two disagreements; reference that cites the notice itself dropped). A Crossref record can describe a container rather than an article, and the case that prompted A4 is exactly that: the Crossref record for 10.1007/s00125-014-3355-0 is titled "Abstracts of the 50th EASD Annual Meeting" and has no authors, so A4's three checks see only one disagreement. A6 therefore adds a fourth check, the notice's own title, when it names what it updates ("Retraction Note: …", "Retracted: …", "Correction to: …"). Two disagreements out of four still reject.

Was the metadata there at the snapshot? Crossref serves current records. For each notice record, `created` (first registration of the notice DOI) and `deposited` (latest deposit) bound when its `update-to` relation could first be seen. The cutoff is 2026-10-07T11:07:45Z, the last PubMed response of the frozen run. A link is `present_at_snapshot` when the notice record was registered and last deposited before the cutoff, `registered_before_redeposited_after` when it was registered before and deposited again after, and `registered_after_snapshot` when the notice DOI did not yet exist.

Comparison. The unit is a unique work and a claim type. As in the study's unique-work count (S4), references labeled ambiguous in the coverage ledger are left out. A work is flagged by PubMed or Retraction Watch when one of its references carries an arm C claim of that type after A4, and by Crossref when one of its references has a kept Crossref link of that type. For every work flagged by at least one source and missed by another, the code assigns a reason for each missing source from the records (`why_not_PM`, `why_not_RW`, `why_not_CR` in `outputs/tables/a6_discordant_works.csv`). For Retraction Watch the reason search also tries a title match (RapidFuzz `token_sort_ratio` ≥ 90) against the snapshot, with a journal check so that a different paper with the same title is not taken for the work.

3. Results

Crossref holds records for 46,846 of the 47,394 works with a DOI; the DOIs of 548 works return 404. On the study DOIs Crossref lists 999 `updated-by` entries, 944 from publishers and 55 from Retraction Watch. For 898 of the 902 publisher entries of the compared types, the notice record's own `update-to` names the work's DOI. Of the other four, two are self-updates and two are *Diabetes Care* errata whose notice records carry no `update-to` although the article records list them. The 1,284 reference-level Crossref links include two that the checks reject: the Ashtagi reference (Padhy et al. record, A4 checks) and the Kramná reference (supplement record, rejected only because the notice subject also disagrees). These are the two false links that A4 removed from the Retraction Watch and PubMed claims.

Every notice record behind a kept link (41 retraction, 11 concern and 848 correction work–notice pairs) was registered and last deposited before the cutoff. On the record dates Crossref gives, the Crossref flags in this comparison were in place when the study's snapshot was taken; none comes from a deposit made after 7 October.

Works (with a DOI, ambiguous references left out)	Retraction	Expression of concern	Correction
Flagged by any source	46	21	1,445
PubMed / Retraction Watch / Crossref	39 / 35 / 39	17 / 13 / 10	1,388 / 4 / 801
All three	26	5	2
PubMed and Retraction Watch, not Crossref	7	4	2
PubMed and Crossref, not Retraction Watch	6	5	742
Retraction Watch and Crossref, not PubMed	2	0	0
PubMed only / Retraction Watch only / Crossref only	0 / 0 / 5	3 / 4 / 0	642 / 0 / 57
Positive agreement, Crossref with PubMed	0.82	0.74	0.68
Positive agreement, Crossref with Retraction Watch	0.76	0.43	0.01
Positive agreement, PubMed with Retraction Watch	0.89	0.60	0.01

Positive agreement is 2 × both / (flags of one + flags of the other). Retraction Watch records few corrections, so its correction figures say more about scope than about agreement.

Crossref adds five retraction-flagged works to the 41 with a DOI that PubMed or Retraction Watch flag; it adds no concern. If Crossref's Retraction Watch entries were counted as well, Crossref's API would show all 46 retraction works, but that would count Retraction Watch twice.

Notice dates. Where a study claim and a Crossref link point to the same notice DOI, the Crossref update date falls on the study's notice date or inside its interval for 21 of 25 PubMed retraction claims and 20 of 21 Retraction Watch retraction claims, 10 of 11 and 5 of 5 concern claims, and 371 of 633 PubMed correction claims. Two kinds of Crossref date cannot stand for a notice date. In 10 retraction links the publisher marked the article's own record as retracted (notice DOI = article DOI), and the update date it deposited is the article's date: 2014-09-01 for a *BBA – Molecular Basis of Disease* paper whose PubMed retraction is dated 25 April 2025, 2023-02-01 for a *Heliyon* paper retracted on 17 March 2026 per Retraction Watch. In 44 links (1 retraction, 43 corrections) the update date is more than a month before the notice DOI existed; the *Nutrients* retraction of 2025 carries the article's 2019 date. These links are marked in `date_flag`, and their dates are not used for timing.

4. Every discordant work

Retraction (20 works)

Crossref only (5):

1. 10.1002/14651858.CD013858, Cochrane review cited by PMC8224679. Crossref type `withdrawal` (Cochrane's withdrawn-version deposit, notice CD013858.pub2, 29 December 2021, after the review). The DOI has no PubMed record in the study and no Retraction Watch record. A Cochrane withdrawal of a version is not a retraction for cause; it is counted because the study's rule maps withdrawals to retractions, as `rae recheck` already did.
2. 10.1007/s00521-021-06431-7, *Neural Computing and Applications*, cited by PMC11983615. Retraction Note of 25 May 2026, after the review. The journal is not in PubMed, and neither the 2026-10-06 Retraction Watch file nor Crossref's current Retraction Watch entries list it. Crossref is the only source that knows of this retraction.
3. 10.1007/s12652-020-02456-3, *Journal of Ambient Intelligence and Humanized Computing*, cited by PMC8831075. Retraction Note of 28 November 2024, after the review. Same situation: no PubMed record, no Retraction Watch record.
4. 10.3402/gha.v6i0.19305, *Global Health Action*, cited by PMC4627942. The publisher deposited a notice titled "Corrigendum" (10.3402/gha.v7.24080, 13 March 2014) with type `retraction`, and the article's Crossref title now begins "RETRACTED:". PubMed records a correction and Retraction Watch record 44075 a correction ("Error in Text") with the same notice DOI. The records conflict; whether the article was retracted was not settled here and is listed for review.
5. 10.1016/S0140-6736(20)30367-6, the 2020 *Lancet* Commission on dementia, cited by three reviews. Crossref lists a retraction because the retracted *Lancet Public Health* item "Addressing hearing loss at all ages" (10.1016/S2468-2667(23)00083-X) carries `update-to` relations to four DOIs, this one among them. PubMed records only a correction. This is a deposit error, not a retraction of the Commission report. The link checks keep it: the Crossref record matches the reference, and only the notice subject disagrees. One disagreement on the notice subject alone also occurs on correct links (for example *European Journal of Heart Failure* notices that correct several articles at once), so it was not made a rejection rule after the fact.

PubMed and Crossref, not Retraction Watch (6). All six are retractions that neither the snapshot nor Crossref's current Retraction Watch entries list, found by DOI, PMID and title: *BMC Family Practice* 2014 (10.1186/1471-2296-15-8; notice 6 February 2025), *BBA – Molecular Basis of Disease* 2014 (10.1016/j.bbadis.2014.05.015; PubMed notice 25 April 2025, self-update in Crossref), *Cardiovascular Diabetology* 2017 (10.1186/s12933-017-0544-4; 11 September 2026), *Nutrients* 2019 (10.3390/nu11102496; 9 May 2025), *Obesity Surgery* 2021 (10.1007/s11695-021-05354-5; 7 May 2026) and *International Wound Journal* 2023 (10.1111/iwj.14507; 21 April 2025). The *International Wound Journal* retraction is the only one dated before its citing review (PMC12352421).

Retraction Watch and Crossref, not PubMed (2):

- 10.1002/14651858.CD001872.pub2 (Cochrane, childhood obesity). Retraction Watch record 30719 and a Crossref `withdrawal` (CD001872.pub3, 7 March 2019) mark the version as superseded; PubMed 19160202 carries an update relation, which the study does not count as a retraction.
- 10.1016/j.heliyon.2023.e13323. Retraction Watch record 70494 (17 March 2026); Crossref self-update. The PubMed record as retrieved on 7 October 2026 has no retraction relation.

PubMed and Retraction Watch, not Crossref (7). In six, the notice DOI is registered at Crossref and names the work only through Crossref's Retraction Watch entry; the publisher deposited no `update-to`: *Family Practice* 2009 (10.1093/fampra/cmp003), PREDIMED in the *New England Journal of Medicine* (10.1056/NEJMoa1200303), *British Journal of Nutrition* 2013 (10.1017/S0007114513002791), *JAMA Psychiatry* 2014 (10.1001/jamapsychiatry.2014.763), *British Journal of Clinical Pharmacology* 2015 (10.1111/bcp.12788) and *BJPsych Open* 2018 (10.1192/bjo.2018.12). In the seventh, the 2021 *Lancet* article 10.1016/S0140-6736(21)01640-8, the publisher deposited the retraction notice 10.1016/S0140-6736(24)00318-0 with type `erratum`, so Crossref counts it as a correction.

Expression of concern (16 works)

- Publisher deposited no relation; Crossref shows only the Retraction Watch entry (6): *European Journal of Clinical Nutrition* 10.1038/ejcn.2012.189 and 10.1038/ejcn.2013.296, *CMAJ* 10.1503/cmaj.140764, *British Journal of Nutrition* 10.1017/S0007114516003457, *Biological Trace Element Research* 10.1007/s12011-017-1056-5 and *Probiotics and Antimicrobial Proteins* 10.1007/s12602-017-9313-7. For four

of them (both *EJCN* papers and the two Springer papers) PubMed, as retrieved on 7 October, also has no concern relation; Retraction Watch is the only source.

- Publisher deposited the concern as another type (2): *Journal of Nutrition* 10.3945/jn.113.177550 (notice 10.1093/jn/nxaa340 deposited as `erratum`) and *Public Health Nutrition* 10.1017/S1368980015000609 (notice deposited as `correction`).
- Notice registered at Crossref without any `update-to` (3): *Hormone and Metabolic Research* 10.1055/s-0042-105441 (notice 10.1055/a-1736-9432) and two *Medicine (Baltimore)* papers, 10.1097/MD.0000000000014918 and 10.1097/MD.0000000000019471 (shared notice 10.1097/MD.0000000000021299). None of the three is in Retraction Watch; PubMed is the only source.
- PubMed and Crossref, not Retraction Watch (5): two *Lipids* papers (10.1007/s11745-014-3901-z, 10.1007/s11745-017-4295-5; Wiley notices of 26 August 2024), *Liver International* 10.1111/liv.12990 (26 August 2024), *Wound Repair and Regeneration* 10.1111/wrr.12537 (20 August 2024) and *PLOS ONE* 10.1371/journal.pone.0292100 (8 May 2025). The snapshot record with the closest title to the *Wound Repair and Regeneration* paper (28443) belongs to a different paper with a near-identical title in *Biological Trace Element Research*.

Correction (1,443 discordant works)

Each work has its reason in `a6_discordant_works.csv`; the counts are in `a6_discordance_reasons.csv`.

- Not in Crossref while PubMed flags it (644 works): the PubMed notice has no DOI (342), the notice is registered at Crossref without `update-to` (294), its `update-to` names another DOI (4), the publisher deposited another type (2), the notice DOI has no Crossref record (1), Crossref shows only a Retraction Watch entry (1).
- Crossref only (57): the PubMed record has no correction relation (38), the work has no PubMed record (10), PubMed carries a relation of another type (9).
- Retraction Watch is missing for 1,441 of the 1,443 works: it holds no record for 1,430 and a record of another nature for 11. Retraction Watch does not aim to cover corrections, so this is a statement of scope.

5. What this changes

Nothing in the study's numbers. A6 is a comparison next to them. For the paper it supports four statements, each traceable to the files below.

- The choice of status source matters as much with three sources as with two. Of 46 retraction-flagged works, 26 are flagged by all three.
- A checker that reads Crossref's publisher deposits as well as PubMed and Retraction Watch would have flagged 47 works in this sample (46 with a DOI and the one without), five more than the study's 42. Two of the five are retractions known to no other source, one is a deposit error, one is an unresolved conflict between a corrigendum and a retraction, and one is a Cochrane withdrawal.
- Retraction Watch's file of 6 October 2026 lacked eight retractions that publishers had deposited at Crossref (six of them also in PubMed), the oldest dated 28 November 2024.
- Crossref's update dates need the same care as the other sources. Self-updated records and back-dated deposits give the article's date, not the notice's.

6. Code, tests, outputs

- `src/rae/crossref_status.py` (retrieval, parsing, deposit classes, link checks) and `src/rae/crossref_analysis.py` (comparison and reasons), both new. No existing module was changed.
- `scripts/deep_2026-10-24/s10_a6_crossref.py`: `fetch-works`, `fetch-notices` (network) and `analyze` (offline). Run from the project root with `PYTHONPATH=src` and `RAE_ROOT` set; with the cache in place, `RAE_OFFLINE=1` reruns everything. Two offline runs of `analyze` gave byte-identical outputs.
- `tests/test_crossref_status.py`: synthetic Crossref messages for source labels and type mapping, the snapshot classes, notice-subject parsing, the supplement-DOI case (kept by A4's three checks, rejected with the notice subject), the same-title case, a correct link, a cited notice, date flags, and batched retrieval with a mock transport that checks the User-Agent and the cache. The full suite passed (81 tests, 24 October 2026).
- Outputs in `outputs/tables/`: `a6_summary.json`, `a6_concordance.csv`, `a6_work_status.csv`, `a6_discordant_works.csv`, `a6_discordance_reasons.csv`, `a6_crossref_updates.csv`, `a6_crossref_links.csv`, `a6_deposit_timing.csv`, `a6_notice_date_pairs.csv`, `a6_notice_date_agreement.csv`. In `data/interim/`: `a6_crossref_works.json.gz`, `a6_crossref_notices.json.gz`, `a6_pubmed_notice_dois.parquet`. In `data/processed/`: `a6_crossref_updates.parquet`, `a6_crossref_links.parquet`.

7. Limits

- `deposited` is the latest deposit date. A relation added and later removed before the cutoff would leave no trace, and the classes rely on Crossref's dates being right. The dates above show that publishers do not always deposit them correctly.
- Crossref covers only DOIs it registers. The 548 works with DOIs from other agencies cannot have a Crossref flag, and works without a DOI were not compared.
- The notice-subject check was added before the data were seen, but it was written with the Kramná case in mind. It changes one decision (that case), and that change is listed in `a6_summary.json`.
- The descriptions of the discordant retraction and concern works come from reading the cached records. The *Global Health Action* conflict and the *Lancet* deposit error were not checked against the publishers' web pages.
- A cross-check with separately written code (`WORK/deep/S10/a6/crosscheck_a6.py`, raw cache files read directly) reproduced the 1,284 links and every pattern count. It was written by the same agent. A re-computation by a separate agent has not yet been done.

Amendment A7: replication on the 6,993 frame records not selected

- Written: 2026-10-24, between 22:06Z and 22:15Z (UTC). The file's SHA-256 is recorded in `validation/ledger_2026-10-24.jsonl`, in the line appended after this file was finished.
- Status: post hoc. Every result of the 1,000-review study was known when this was written: the manuscript submitted on 18 October 2026, the item-level AI check of 24 October 2026, amendments A4 (with its rerun), A5 and A6. No data on the 6,993 records described here had been retrieved. Before writing, the agent sent six requests to time the endpoints (21:59:52Z to 22:00:00Z). None of them concerned the 6,993 records: one PubMed EFetch of 200 PMIDs cited in the main study, one ID Converter call for 200 cited PMCIDs from the main study, one DOI search for 40 DOIs from the main study, and the PMC cloud listing, metadata and XML of PMC8518965, a main-study review. The replication root and its manifest were created at 22:05:03Z without any network request.
- Drafted by an AI agent (Claude Opus 5.5, claude-opus-5-5, via Claude Code) for the author. The author has not yet reviewed it; the review is queued in the deep-pass author review list.
- `docs/analysis_plan.md`, `config/config.yaml`, amendments A1 to A6, `src/rae/` and `sql/` are not edited. A7 changes no number of the main study, whatever it finds.

1. Why

The main study is one hash-ordered draw of 1,000 records from a frame of 7,993. Its retraction counts are small: 44 flagged reference occurrences in 40 reviews, and 7 occurrences in the A5 key figure (a non-replacement retraction whose notice came at least 183 days before the review went online). Intervals around counts this small are wide, and a reader can fairly ask whether the pattern would hold in another draw. The other 6,993 records of the same frame are that draw, and they cost only retrieval time.

There is a second reason. The A4 link guard was written after its trigger (the Diabetologia supplement DOI) had been seen, and its title threshold was set on the main study's links. The replication is the first set of links it meets that played no part in writing it.

2. Population

The 6,993 rows of the frozen frame manifest (`data/interim/frame_manifest.csv`, SHA-256 `4ae5614c…6b5b`, frozen 2026-10-07T10:02:46Z) with `selected = False`. The replication manifest, `replication/A7_2026-10-24/data/interim/frame_manifest.csv` (SHA-256 `8d70976c…475d`), has the same rows, order and columns, with `selected` flipped and nothing else changed (`replication/A7_2026-10-24/frame_manifest_A7_derivation.json`). As in section 3 of the plan, records are not replaced, and records that cannot be retrieved or parsed are reported by reason.

3. What runs

- `rae acquire`, `rae build` (which applies the A4 guard), `rae analyze` and `rae a5`, called through `scripts/deep_2026-10-24/s10_a7_replication.py run` with `--config config/config.yaml` and `RAE_ROOT=replication/A7_2026-10-24`. The hashes of every file in `src/rae/` and `sql/`, of the config and of the driver are in the ledger line.
- Differences from the 7 October run, none of them in the code:
  - Retrieval date. PubMed records, ID Converter answers and PMC cloud files are retrieved on 24 and 25 October 2026, not 7 October. The Retraction Watch file is the same pinned snapshot (data for 6 October 2026, SHA-256 `063df713…db4e5`).
  - User-Agent `[affiliation]-deep-pass/1.0`. The driver gives it to the same `CachedClient` with the same settings: 0.5 s between NCBI requests (at most 2 a second, below NCBI's limit of 3 without a key), 5 attempts with back-off, 200-record EFetch batches, 40-DOI searches, 4 workers for the PMC cloud. No contact address is sent.
  - A cache of its own under the replication root, empty at the start, so the main study's cache and request logs are not touched.
- Not run: `rae validate`, `render`, `package` and `recheck`, and the A6 Crossref comparison. A6 on this set would need roughly 10,000 Crossref requests at one a second, about three more hours; it is left for later.

4. Expected load

Estimated before the run from the main study's 1,000 reviews (unique identifiers accumulated over random review orders and extrapolated with a power-law fit, 20 orders), the 7 October request logs and the 24 October probe. Script and output: `WORK/deep/S10/replication/feasibility_estimate.py` and `feasibility_estimate.json`.

Item	Central	Upper (linear growth)
Unique cited PMIDs	282,424	330,412
Unique cited PMCIDs	129,306	149,727
NCBI requests (EFetch, ID Converter, ESearch)	2,519	2,931
of which EFetch of cited records, 200 per request	1,413	1,653
PMC cloud requests (listing, metadata, XML)	21,077	21,077
Data received, uncompressed	6.9 GB	7.8 GB
Wall-clock time, all four stages	2.65 h	2.92 h

The long step is EFetch of the cited records: each 200-record response took 3.5 s in the probe, so NCBI traffic averages about 0.4 requests a second. Each response is cached with its SHA-256, so an interrupted run restarts from the cache.

5. Declared measures

Each is read from the tables the pipeline writes, in the same way for both runs. The main-study values are those after A4.

Key	Measure	Grain	Main study
M01	Any usable DOI, PMID or PMCID	occurrence	92.2% (57,828 / 62,740)
M02, M03	DOI; PMID	occurrence	89.8%; 89.4%
M04 to M07	Ledger: one verified PubMed record; DOI not in PubMed; ambiguous; no identifier	occurrence	88.9%; 2.7%; 0.5%; 7.8%
M08 to M11	Retraction; expression of concern; correction; any status claim (arm C)	occurrence	44; 24; 2,165; 2,260 of 62,740
M12 to M15	Reviews with a retraction; concern; correction; any claim (arm C)	review	4.0%; 1.7%; 79.6%; 80.7%
M16	Unique works with a retraction claim	work	42 / 48,691
M17	Notice before the review went online (primary rule)	retraction-flagged occurrence	14 / 44
M18	Correction timing determinate	correction-flagged occurrence	64.0% (1,386 / 2,165)
M19 to M21	Both sources; PubMed only; Retraction Watch only	retraction-flagged occurrence	34; 7; 3 of 44
M22, M23	Non-replacement retraction, notice at least 183 days before the review (A5)	review; occurrence	7 / 1,000; 7 / 62,740
M24, M25	Retract and replace; Hindawi DOI prefix 10.1155 (A5)	retraction-flagged occurrence	7 / 44; 10 / 44

Also reported, without intervals: the cohort flow, references per review, the arm increments (what the crosswalk and PubMed add), the A4 guard's links checked and rejected by rule, and the number of PubMed claims whose notice date falls after 7 October 2026 (relations the main study could not have seen).

6. Comparison

- Each replication estimate carries the interval the pipeline already gives: the review-level cluster bootstrap (2,000 resamples, seed 20261007) for occurrence shares and Wilson intervals for review shares, work shares and shares within flagged occurrences. M17 and M19 to M25 take Wilson intervals computed by the driver from the pipeline's counts; for M23, an occurrence share, that interval ignores clustering by review.
- The difference, replication minus main, gets a 95% interval by the MOVER method (Zou and Donner, Stat Med 2008; Newcombe, Stat Med 1998), built from each estimate's own interval. The two sets are disjoint parts of one fixed frame. For a difference of means, treating them as independent samples gives essentially the same variance as the finite-population calculation, so no correction is applied.
- There are 25 intervals and no adjustment for multiplicity. One or two excluding zero by chance would not be surprising. No measure is a pass or fail test.
- Frame-wide values (main plus replication, the whole frame apart from records that could not be used) are given as ratios of summed counts, without intervals. Unique works are counted within each run and are not

pooled.

7. The guard on new links

Every link the A4 guard rejects in the replication is listed in `replication/A7_2026-10-24/outputs/tables/amendment_a4_rejected_links.csv`, and every link kept with one disagreeing check in `amendment_a4_review_list.csv`. The agent reads each rejected retraction or concern link, and each retraction or concern link kept with one disagreement, against its matched record and says whether the decision looks right. The guard is not changed in response. A false link that the guard kept is reported and counted, not removed.

8. Fallback

If `rae acquire` has not finished by 2026-10-25 02:30Z, or NCBI refuses the requests, the full run stops. The pipeline then runs on 1,068 of the 6,993 records: those with the smallest SHA-256 of `"A7-fallback|20261024|" + PMID`, with seed string `A7-fallback|20261024|` recorded here. 1,068 is the simple-random-sample size for a 95% interval half-width of at most 3 percentage points on any review-level share (worst case p = 0.5, no finite-population correction). Occurrence-level shares come out narrower at that size; the main study's 1,000 reviews gave ±0.7 points for identifier coverage. Cached PMC cloud responses would be reused.

9. Where the results go

The run writes its tables, figures and workbooks under `replication/A7_2026-10-24/outputs/`. The comparison is written to `outputs/tables/a7_replication_comparison.csv` and `outputs/tables/a7_summary.json`. Stage times and exit codes are in `replication/A7_2026-10-24/run_status.json`, and the logs in `replication/A7_2026-10-24/logs/` and `.local/logs/`. A second count of the key numbers, written separately from the pipeline, is saved in `validation/`. The write-up is `WORK/deep/S10/replication.md`.

10. Limits

- Same frame, same filters. The replication speaks to sampling variation within one PubMed-defined frame of open-access reviews from 2015 to 2025. It says nothing about other topics or about reviews outside PMC's open-access subset.
- Seventeen days later. PubMed relations added between 7 and 24 October are visible only to the replication. They are counted (section 5) so their effect can be seen.
- Written after the main results were known. The measures follow the manuscript's headline numbers and the A5 key figure. This file and its ledger line are internally time-stamped; that does not make the replication preregistered.

Addendum A7.1: the A5 split when a flagged reference has no single work

- Written: 2026-10-25, between 00:29Z and 00:35Z (UTC). The file's SHA-256 is recorded in `validation/ledger_2026-10-24.jsonl`, in the line appended after this file was finished.
- Status: post hoc deviation from amendment A7 (`docs/amendment_A7_2026-10-24.md`, SHA-256 `8d4d0597…2c98`). Written after `rae acquire`, `rae build` and `rae analyze` had finished on the replication and after `rae a5` had stopped with an error. When this was written the agent had seen the replication's claim counts by type, the A4 guard's link decisions and the error message, but no A5 table for the replication, because none had been written.
- Drafted by an AI agent (Claude Opus 5.5, claude-opus-5-5, via Claude Code) for the author, who has not yet reviewed it.

What happened

At 00:27:26Z `rae a5` stopped with `ValueError: retraction-flagged occurrence without a work` for six references: PMC11949964 B71, PMC12206637 B17, PMC6463004 B25, PMC6644532 B22, PMC8002607 B56 and PMC8406725 CR40. All six are labeled ambiguous in the coverage ledger, so they have no work id. A5 then tries the PubMed record named by the reference's retraction claims; that works only when the claims name exactly one record that is itself a work. Here they name two records each, or one that is not a work key: two versions of the Cochrane review CD001872 (2003 and 2009, cited by two of the six), two papers with the same title and authors in different journals (one reference), two references that print an identifier of the retraction notice next to the article's, and one reference whose printed DOI is an issue-level DOI, so that the one record its claims name is not a work key. The main study had one such reference (PREDIMED, PMC7551578 B59), which the single-record rule resolved; the replication's six are the first that it cannot.

The error is a deliberate stop in A5's code, not a crash. A7 said `rae a5` runs unchanged, so a deviation is needed to report the A5 measures at all.

The deviation

A5's own function (`rae.amendment_a5.build_tables`) runs unchanged. In an in-memory copy of the replication ledger, each of the six references gets the work key `occurrence:<occurrence_id>`. Each then counts as a work of its own, and its notice type (retraction, or retract and replace) comes from its own claims only. The six are labeled in the `work_key_source` column of `amendment_a5_retraction_occurrences.csv`. The script is `scripts/deep_2026-10-24/s10_a7_1_a5_occurrence_keys.py`. It first checks that its copy of A5's key rule picks the same six references that the error names, and it refuses to run if `rae a5` would work without it. Outputs go to the replication root only; the main study's A5 files and code are not touched.

What it can and cannot change

- Occurrence and review counts (A7 measures M22 to M25) depend on each reference's own claims and timing. The work key changes them only through the rule that a replacement recorded on one reference of a work applies to every reference of that work. A reference keyed by itself cannot inherit a replacement from another reference, and cannot pass one on.
- A5's work count can be too high by up to six, if any of the six cites a work that another flagged reference also cites. A7's declared work measure (M16) comes from `rae analyze`, not from A5, and is not affected.
- The replication's A5 results are reported with this note wherever they appear.